Mid Sem-1 (BEE – RAPC202)

Max. Time- 50 min. Max. Mark-15

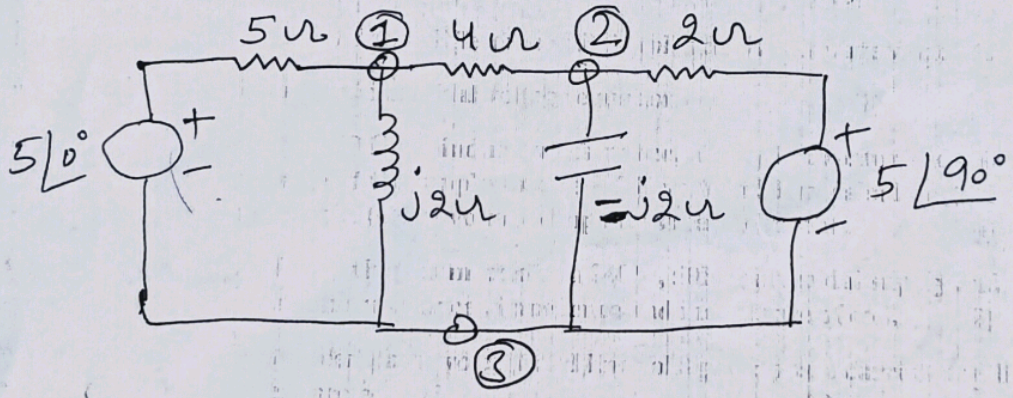
Q1. What is R.M.S value of an A.C quantity. Obtain expression for the r.m.s value of a sinusoidal current in terms of its maximum value. (3)

Q2. Define (i) Time period (ii) Phase difference (iii) Amplitude (iv) Average value (2)

Q3. Write the different forms of a sinusoidal voltage equation . (1)

Q4. A series circuit consists of 300 Ω noninductive resistance, a 7.95μf capacitor and 2.06 h inductor of negligible resistance. If the supply voltage is 250 volts at 50 hz calculate (a) The circuit current (b) the phase angle (c) the voltage drop across each element . (4)

Q5. A coil takes a current of 2A , when connected to a 240V , 50hz sinusoidal supply and consumes 200 watts. Calculate the resistance , impedence and inductance of coil. (2)

Q6. Write the relationship between phase voltage and line voltage & phase current and line current in three phase star connected load. (1)

Q7. For the given circuit write the nodel matrix in the form [Y] [V] = [I] for determining the nodel voltages. (2)